AMOD5430/COIS3510 Data Visualization – Workshop 2 Instructions

Configure Your Environment

Please follow these instructions to set up your computer environment:

- Install R and RStudio on Windows
- Install R and RStudio on Mac
- Install R and RStudio on Ubuntu

Alternatively, you may set up a Python-based environment:

- Installing Python on Windows
- Installing Python on Mac
- Installing Matplotlib on Windows
- Installing Matplotlib
- Installing Matplotlib on Mac
- Installing Jupyter on Windows
- Installing Jupyter on Mac

Displaying Multivariate Data in R

Open RStudio and complete the following plots by entering code snippets below. Please submit a PDF file containing the charts you generated, and the rmd file, to the Blackboard assignment.

Basic Scatter Plot

```
# Basic scatter plot
x <- c(6,7,8,7,2,3,9,4,10,15,9,7)
y <- c(98,85,87,88,112,104,87,94,78,78,85,86)
plot(x,y)
```

Scatter Plot with Multiple Classes

```
# scatter plot with multiple classes
plot(iris$Petal.Length, iris$Petal.Width, pch=c(23,24,25)[unclass(iris$Species)], main="Iris Data")
```

Scatter Plot with 4 Dimensional Data

```
# Scatter plot with 4 dimensions (petal lenght added)
attach(iris)
plot = plot(Sepal.Length,Sepal.Width,type="p",
             cex = Petal.Length,
             bg = Species, # Background color
             pch = 21)
```

Scatter Plot Matrix

```
# scatter plot matrix
pairs(~Sepal.Length+Sepal.Width+Petal.Length+Petal.Width,data=iris,
       upper.panel = NULL)
```

Scatter Plot Lattice

```
# scatter plot lattice
library("lattice")
xyplot(Sepal.Length ~ Petal.Length | Species,
        group = Species,
        data = iris,
        type = c("p", "smooth"),
        scales = "free")
```

Parallel Axes

```
# parallel axes
library(GGally)
ggparcoord(data,
            columns = 1:4, groupColumn = 5
 )
```

Chernoff Faces

```
# Chernoff Faces
install.packages("aplpack")
library("aplpack")


# View the faces

faces(face.type=1)

# View the effect of the variables

faces(rbind(1:3,5:3,3:5,5:7))
```

```
# View the data

data(longley)
faces(longley[1:9,],face.type=0)


# Using in a scatter plot

plot(longley[1:16,2:3],bty="n")
a<-faces(longley[1:16,],plot=FALSE)
plot.faces(a,longley[1:16,2],longley[1:16,3],width=35,height=30)
```

Radar Charts

```
# Basic chart

install.packages("fmsb")
library("fmsb")
df <- data.frame(rbind(rep(10, 8), rep(0, 8),
                       matrix(sample(0:10, 8),
                              nrow = 1)))


colnames(df) <- paste("Var", 1:8)
radarchart(df)


# Customize

radarchart(df,
            cglty = 1,       # Grid line type
            cglcol = "gray", # Grid line color
            cglwd = 1,       # Line width of the grid
            pcol = 4,        # Color of the line
            plwd = 2,        # Width of the line
            plty = 1)        # Line type of the line


# Multiple groups or classes

set.seed(1)
df2 <- data.frame(rbind(rep(10, 8), rep(0, 8),
                        matrix(sample(0:10, 24,
                                      replace = TRUE),
                               nrow = 3)))

colnames(df2) <- paste("Var", 1:8)
radarchart(df2,
            cglty = 1,       # Grid line type
            cglcol = "gray", # Grid line color
            pcol = 2:4,      # Color for each line
            plwd = 2,        # Width for each line
            plty = 1)        # Line type for each line


# Fill the areas

areas <- c(rgb(1, 0, 0, 0.25),
```

```
            rgb(0, 1, 0, 0.25),
            rgb(0, 0, 1, 0.25))
radarchart(df2,
           cglty = 1,       # Grid line type
           cglcol = "gray", # Grid line color
           pcol = 2:4,      # Color for each line
           plwd = 2,        # Width for each line
           plty = 1,        # Line type for each line
           pfcol = areas)   # Color of the areas
```

Displaying Multivariate Data in Python/Jupyter

Start your Jupyter Lab and complete the following plots by entering code snippets below. Please submit a PDF file containing the charts you generated, and the Jupyter notebook source file, to the Blackboard assignment.

```
import matplotlib.pyplot as plt

import matplotlib.cm

import seaborn as sns

import pandas as pd

from pandas.plotting import parallel_coordinates

from ChernoffFace import *

import numpy as np
```

Basic Scatter Plot

```
# Define the data

x = [6, 7, 8, 7, 2, 3, 9, 4, 10, 15, 9, 7]

y = [98, 85, 87, 88, 112, 104, 87, 94, 78, 78, 85, 86]

# Create the scatter plot

plt.scatter(x, y)
```

```
# Display the plot
plt.show()

##### Scatter Plot with Multiple Classes

# Load the iris dataset
iris = sns.load_dataset('iris')

# Create a scatter plot with different species as different classes
sns.scatterplot(x='petal_length', y='petal_width', hue='species', style='species', data=iris)

# Add title
plt.title('Iris Data')
plt.show()

##### Scatter Plot with 4 Dimensional Data

# Create a scatter plot with Sepal.Length vs Sepal.Width
# cex equivalent (point size) is Petal.Length and bg equivalent (background color) is Species
sns.scatterplot(x='sepal_length', y='sepal_width', hue='species', size='petal_length', data=iris, sizes=(20, 200), style='species', palette='viridis')

# Add title and show plot
plt.title('Iris Data - Sepal Dimensions with Petal Length as Size')
plt.show()

##### Scatter plot Matrix
```

```
# Create a scatter plot matrix
sns.pairplot(iris, diag_kind='hist')

# Show the plot
plt.show()

##### Scatter Plot Lattice

# Create a faceted scatter plot lattice with a smoothing line
g = sns.lmplot(x='petal_length', y='sepal_length', col='species', hue='species', data=iris,
               height=4, aspect=0.7, scatter_kws={'s': 50}, line_kws={'color': 'black'}, ci=None)

# Customize the scales and titles
g.set_axis_labels('Petal Length', 'Sepal Length')
g.set_titles('Species: {col_name}')

# Show the plot
plt.show()

##### Parallel Axes

# Load the iris dataset
url = "https://raw.githubusercontent.com/vincentarelbundock/Rdatasets/master/csv/datasets/iris.csv"
iris = pd.read_csv(url)

# Parallel coordinates plot
plt.figure(figsize=(10, 6))
parallel_coordinates(iris, 'Species', cols=['Sepal.Length', 'Sepal.Width', 'Petal.Length', 'Petal.Width'],
color=['r', 'g', 'b'])
```

```
# Add title and show the plot
plt.title('Parallel Coordinates Plot - Iris Data')
plt.xlabel('Features')
plt.ylabel('Values')
plt.show()

##### Chernoff Faces
##### Things to run
### - pip install ChernoffFace
### - ModuleNotFoundError: No module named 'pkg_resources' --> pip install setuptools

# Load the Longley dataset from the URL
url = "https://github.com/vincentarelbundock/Rdatasets/raw/master/csv/datasets/longley.csv"
data = pd.read_csv(url)

# Generate Chernoff faces with the Longley data
fig = chernoff_face(data=data,
                    titles=[str(x) for x in list(range(len(data)))],
                    color_mapper=plt.cm.Pastel1)

# Display
fig.tight_layout()
plt.show()

##### Radar Charts

# Data
labels = ['Var 1', 'Var 2', 'Var 3', 'Var 4', 'Var 5', 'Var 6', 'Var 7', 'Var 8']
```

```
num_vars = len(labels)

# Create the data for the radar chart
data = [np.random.randint(0, 11) for _ in range(num_vars)]
data += data[:1]  # Close the loop by repeating the first value

# Define the angle for each axis (in radians)
angles = np.linspace(0, 2 * np.pi, num_vars, endpoint=False).tolist()
angles += angles[:1]  # Complete the loop by appending the first angle

# Customize the chart (similar to R's grid line type, colors, and line width)
fig, ax = plt.subplots(figsize=(6, 6), subplot_kw=dict(polar=True))

# Grid lines
ax.spines['polar'].set_visible(True)
ax.grid(color='gray', linestyle='-', linewidth=1)

# Fill the area under the radar chart
ax.fill(angles, data, color='blue', alpha=0.25)

# Plot the outline of the radar chart
ax.plot(angles, data, color='blue', linewidth=2, linestyle='-')

# Remove the radial labels (y-axis labels)
ax.set_yticklabels([])

# Set the ticks (labels around the radar chart)
ax.set_xticks(angles[:-1])
ax.set_xticklabels(labels)
```

```
plt.show()
```